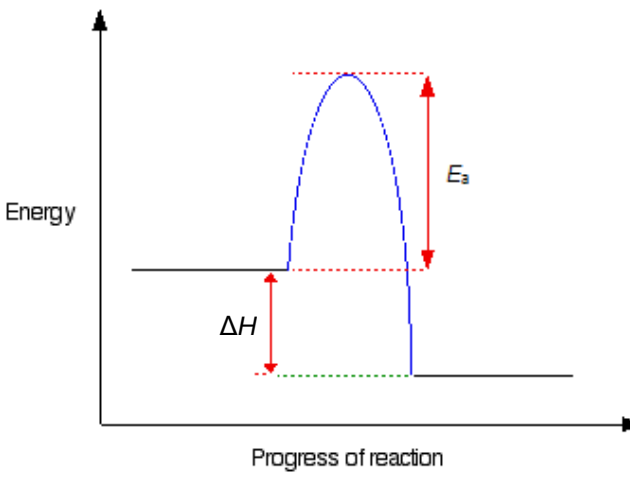


Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
9.	(a)	(i)		(average) energy required to break / released by making (1) 1 mol of bonds (1)	2			2		
		(ii)		bonds broken $\text{C}\equiv\text{C} + 2(\text{H}-\text{H}) = 839 + 2(436) = 1711$ (1) bonds formed $(\text{C}-\text{C}) + 4(\text{C}-\text{H}) = 348 + 4(413) = 2000$ (1) $\Delta H = 1711 - 2000 = -289 \text{ kJmol}^{-1}$ (1) ecf possible		3		3	1	
		(iii)		 products at lower energy than reactants (1) ecf possible from part (ii) ΔH and E_a labelled (1)	2			2		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)			Hess' cycle drawn with all arrows in correct direction / $\Delta H = \Delta H \text{ reactants} - \Delta H \text{ products}$ (1) correct values -1300, -572 and -1600 on arrows / $\Delta H = [-1300 + (-572)] - (-1600)$ (1) $\Delta H = -272 \text{ kJmol}^{-1}$ (1)		3		3	1 1	
	(c)			bond enthalpy terms are averages / same bond in different environment will have slightly different bond enthalpy			1	1		
	(d)			(electrophilic) addition / hydrogenation	1			1		
				Question 9 total	5	6	1	12	3	0